VARSITY PATH

USER STORIES & TASKS

1. Purpose of this document

This document translates the VarsityPath product requirements and the tasks recorders into user stories, acceptance criteria and actionable development tasks. It is intended to help the team understand what value each feature should deliver, how the team can check that it works, and how the work relates to the existing sprint structure.

2. Product context

VarsityPath is an AI-supported university-readiness platform for Grade 11–12 learners. It connects academic results, learning insights, study goals, programme discovery and application tracking in one learner-centred workflow. AI can assist with extracting and explaining information, but learners must review extracted marks before saving, and programme eligibility must be determined using structured institution-specific rules.

3. User stories and acceptance criteria

US-01 — Learner records academic results

Priority: High

As a Grade 11–12 learner, I want to enter and manage my academic results, so that I can understand my performance and use genuine results to guide my learning.

Acceptance criteria

- The learner can enter a subject, topic, mark and assessment date for Improve results.
- Marks are validated within the supported range of 0–100; invalid values receive a clear message.
- The learner can view, edit and delete saved results.
- Saved records belong to the authenticated learner and persist after signing out and back in.
- The dashboard uses saved results only; when there are no results, a clear empty state appears instead of fabricated charts or marks.
- Improve records remain separate from Apply results.

Tasks

- Design the academic results form and results list.
- Implement subject, topic, mark and date fields with validation.
- Connect create, read, update and delete operations to the learner’s Firestore data path.
- Add loading, validation, success and error feedback.
- Test valid marks, invalid marks, edits, deletions, persistence and empty-account behaviour.

US-02 — Learner reviews academic strengths

Priority: High

As a learner, I want to see strengths, weaker subjects and knowledge gaps based on my saved results, so that I know what to focus on when studying.

Acceptance criteria

- Insights are calculated or displayed only when genuine, saved results support them.
- The learner can understand the explanation of strengths and areas needing improvement in plain language.
- Topic-level gaps are shown only when relevant topic data exists.
- The learner can trace insights back to their own saved results.
- If results are missing, the module explains what the learner needs to add rather than displaying invented insights.

Tasks

- Define how results are grouped by subject and topic for insight summaries.
- Implement strengths, weaker-subject and knowledge-gap summaries based on available data.
- Design the insight and empty-state layouts.
- Verify that displayed metrics update after results are added, edited or deleted.
- Test the explanations against representative real or deliberately entered test cases.

US-03 — Learner creates a study plan

Priority: High

As a learner, I want to create study goals and planner tasks, so that I can work systematically toward my target marks and monitor my progress.

Acceptance criteria

- The learner can create a goal with a target mark and relevant due date where supported.
- A study task can include title, description, subject, optional topic, priority, status and estimated duration.
- The learner can mark tasks complete, edit them and reschedule them.
- The planner can show weekly progress based on the learner’s saved tasks.
- Generated or suggested tasks are grounded in actual Improve results, knowledge gaps and active goals.
- Duplicate generation of the same task is prevented, and an empty state appears when no goals or tasks exist.

Tasks

- Create the goal form and goal-management operations.
- Build task creation, editing, completion and rescheduling.
- Add priority, status, subject, due date and duration fields where applicable.
- Connect planner suggestions to actual learner results and knowledge gaps.

- Add duplicate prevention and weekly progress display.
- Test persistence, status updates, empty states and rescheduling.

US-04 — Learner explores university programmes

Priority: High

As a learner, I want to explore programmes at selected Western Cape universities, so that I can identify tertiary options that match my interests and academic profile.

Acceptance criteria

- The explorers include UCT, Stellenbosch University, UWC and CPUT.
- The learner must have at least seven unique Apply subjects before programme discovery is enabled, as specified in the system documentation.
- The explorer presents programme information and relevant admission requirements in a clear format.
- Programme information is based on official sources, with the source and verification date recorded where available.
- The interface does not display calculated scores or programme recommendations when Apply results are absent.
- The learner is told that meeting minimum requirements does not guarantee admission.

Tasks

- Identify programme catalogue fields needed for institution, programme, qualification, faculty and official URL.
- Gather programme information and requirements from official institutional sources.
- Record the scoring method, subject thresholds, additional requirements, academic-year version and last-verified date where available.
- Build university and programme browsing, details and filter views.
- Implement the seven-unique-subject discovery gate and empty state.
- Test programme browsing and verify that sources and limitations are visible.

US-05 — Learner checks programme eligibility

Priority: High

As a learner, I want to compare my Apply results with a programme’s entry requirements, so that I can understand which criteria I meet and which need attention.

Acceptance criteria

- Eligibility uses institution- and programme-specific scoring methods rather than one universal APS formula.
- The result explains the score or minimum where applicable, required subjects, subject thresholds and additional selection requirements.
- The learner can see which criteria are met, unmet or cannot be evaluated because information is missing.

- Changes to Apply marks trigger recalculation using the relevant rule set.
- When a requirement cannot be evaluated reliably, the system discloses the limitation rather than guessing.
- Official source information and verification dates are shown where available; eligibility is not presented as a guarantee of acceptance.

Tasks
- Define a structured representation for institution- and programme-specific rules.
- Implement the correct scoring and eligibility method for each supported programme.
- Compare Apply subject marks against required subjects and thresholds.
- Show satisfied, unmet and unevaluated criteria with clear explanations.
- Test sufficient and insufficient marks, missing subjects, Mathematics versus Mathematical Literacy, and applicable AND/OR conditions.
- Test recalculation when Apply results change and verify the displayed source information.

US-06 — Learner tracks university applications

Priority: High

As a learner, I want to save selected programmes and update their application progress, so that I can keep track of the next steps for each university application.

Acceptance criteria
- The learner can save an institution, programme, application year, status and notes.
- Suggested statuses include planning, not started, in progress, submitted, under review, accepted, not accepted/rejected and waitlisted.
- Verified deadlines and official portal links are used where available; deadlines are not invented.
- Duplicate tracking of the same programme for the same application year is prevented.
- A VarsityPath reference number is labelled as an internal platform reference, not an official university student number.
- The system only changes the status to Submitted after learner confirmation and successful saving, and it explains whether this means submission in VarsityPath or to the university.

Tasks
- Create the application record structure and tracker interface.
- Implement adding, editing and removing tracked programmes.
- Implement the agreed application status options and notes.
- Add duplicate checks for programme and application year.
- Display verified deadlines and official portal handoff links where available.
- Generate or retrieve a unique Varsity Path reference were implemented and label it accurately.
- Test status changes, persistence, duplicate handling and Finish Application messaging.

US-07 — Learner uploads results for AI extraction

Priority: High

As a learner, I want to upload a school results document and review the extracted information, so that I can enter results more efficiently without losing control over their accuracy.

Acceptance criteria

- The described supported formats are PDF, PNG, JPG and JPEG, with a maximum file size of 20 MB.
- The AI may extract subject names, topics, marks and dates, but extracted rows remain provisional.
- The learner can review, edit, remove or add rows before confirming the save.
- No extracted results are saved automatically without learner review and confirmation.
- Unreadable or incomplete documents produce a clear error or review state rather than invented marks.
- AI-service and network failures produce useful recovery guidance.

Tasks

- Build the document upload interface and validate supported file types and size.
- Connect the configured Gemini-based extraction service.
- Map extracted subject, topic, mark and date values into editable rows.
- Implement edit, remove, add and confirm-save actions.
- Keep extracted data provisional until confirmation.
- Test valid, oversized, unsupported, unreadable and incomplete documents, including AI-service failures.

US-08 — Learner asks VarsityPath Assistant for guidance

Priority: Medium–High

As a learner, I want to ask VarsityPath Assistant questions about the platform and my saved information, so that I can understand my next steps more easily.

Acceptance criteria

- The assistant uses the configured Gemini or approved AI service and shows a useful error if the service is unavailable.
- Personal answers use the authenticated learner’s own saved data and respect access controls.
- The assistant does not fabricate marks, application numbers, application statuses, programme requirements or deadlines.
- Scores, eligibility and application status are retrieved from deterministic services or saved records; AI explains these results rather than deciding them.
- The assistant distinguishes general guidance from verified official university information.
- If relevant learner data is missing, the assistant says what is missing and suggests a valid next step.

Tasks

- Build the assistant chat interface and connection to the configured AI service.
- Provide controlled access to the authenticated learner’s relevant saved information.
- Retrieve application numbers and statuses from saved records instead of generating guesses.
- Create response guidance for uncertainty, missing information and official-source limitations.
- Handle loading, unavailable-service and network-error states.
- Test personal data access, missing data, status questions and questions about admission requirements.